

Fractional Structure and Möbius Transformation — Part III: Operational Invariants, Fractional Flows, and Measurement Protocols

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August 16, 2025

Abstract

Parts I–II located the hinge of reciprocal symmetry at $x = 1$ (the “critical-one” hypothesis), with mirror action $I(x) = 1/x$, logarithmic coordinate $s = \log x$, invariant “energy” $E(x) = (\log x)^2$, and a half-twist normal form around $x = 1$. **This Part III** turns the hypothesis into *operational content*: (i) we package observables into an invariant pair (E, ϕ) with $\phi(s) = \text{sign}(s)$; (ii) we introduce continuous *fractional flows* that realize the half-twist on the projective line; and (iii) we give *measurement protocols* and falsifiable tests that do not depend on units.

Keywords Möbius transformation; reciprocal symmetry; operational invariants; projective line; half-twist; measurement protocol; fractional flow.

1 Introduction & aim

On $\mathbb{R}_{>0}$ let $I(x) = 1/x$. In $s = \log x$, the action is the reflection $s \mapsto -s$. Following Parts I–II we keep

$$\rho(x) := |\log x| = |s|, \quad E(x) := \rho(x)^2 = (\log x)^2,$$

so E is I -invariant and minimized uniquely at $x = 1$. We now add a *phase bit*

$$\phi(x) := \text{sign}(\log x) \in \{-1, +1\},$$

which flips under I . The pair (E, ϕ) separates “magnitude” from “side” and will serve as an operational fingerprint of reciprocal phenomena.

2 Operational invariants (E, ϕ)

Properties. (i) E is scale-free and I -invariant: $E(Ix) = E(x)$. (ii) ϕ is the twist phase: $\phi(Ix) = -\phi(x)$. (iii) Small deviations $x = 1 \pm \varepsilon$ give $E \simeq \varepsilon^2$ and $\phi = \pm 1$.

Odd/even response test. For an observed response $F(x)$ driven by scale, define

$$F_{\text{even}}(x) = \tfrac{1}{2}[F(x) + F(1/x)], \quad F_{\text{odd}}(x) = \tfrac{1}{2}[F(x) - F(1/x)].$$

Then F_{even} depends only on E (insensitive to side), while F_{odd} carries the phase ϕ . Empirically, odd parts should flip across the mirror.

3 Fractional flows and the continuous half-twist

On the projective line $\mathbb{P}^1$, introduce the hinge coordinate

$$u = \frac{x-1}{x+1} \in (-1, 1), \quad x = \frac{1+u}{1-u}.$$

(A) Geometric rotation (exact half-twist). Set $u = \sin \theta$ and drive a uniform rotation $\dot{\theta} = \omega$. Then

$$u(t) = \sin(\theta_0 + \omega t), \quad T = \pi/\omega \Rightarrow u(T) = -u(0).$$

This gives a continuous realization of the discrete symmetry $u \mapsto -u$ in finite time. Along this flow $E = (\log x)^2$ is π -periodic in θ and minimized at $\theta \equiv 0 \pmod{\pi}$, i.e. $x = 1$.

(B) $\text{PSL}(2, \mathbb{R})$ lift (one-parameter Möbius subgroup). Define

$$M_t(x) = \frac{x+t}{1-tx}, \quad t = \tan \theta, \quad M_t \circ M_s = M_{\frac{t+s}{1-ts}}.$$

In the limit $t \rightarrow \infty$ one obtains $x \mapsto -1/x$; on the projective line the sign is unobservable, so this implements the mirror I projectively. Use (A) when sign matters physically, and (B) as an algebraic homotopy within $\text{PSL}(2, \mathbb{R})$.

4 Measurement protocols

Protocol A (energy invariance). Calibrate “unity” so the neutral point is $x = 1$. Sample symmetric pairs (x, e^s) and $(1/x, e^{-s})$ and verify

$$E(e^s) = E(e^{-s}) = s^2.$$

Deviations quantify calibration error rather than physics.

Protocol B (phase flip). Measure a response F . Check flipping:

$$F(e^s) \stackrel{?}{=} -F(e^{-s}) \Rightarrow F \text{ encodes the phase } \phi.$$

If instead $F(e^s) = F(e^{-s})$ then F is energy-like.

Protocol C (continuous half-twist). Drive the system along a controlled one-parameter path in the hinge coordinate, e.g. $\theta(t) = \theta_0 + \omega t$ (implemented as a projective gain schedule in x). Predict a sign inversion of all odd responses at $t = \pi/\omega$ with E unchanged.

5 Re-reading 0 and ∞ (operational view)

In Euclidean arithmetic division by 0 is undefined; in the reciprocal geometry 0 and ∞ are a dual pair exchanged by I and sit at equal “energy” $E \rightarrow +\infty$. Operational singularity thus manifests not at 0 but at the hinge 1, where orientation reverses and the twist accumulates.

6 Minimal schema (math $\leftrightarrow$ philosophy)

- **Singularity at unity.** $x = 1$ fixed with $I'(1) = -1 \leftrightarrow$ hinge of duality.
- **Invariant energy.** $E = (\log x)^2 \leftrightarrow$ auditable neutrality (strength independent of side).
- **Half-twist act.** Discrete $u \mapsto -u$ and its continuous realization $\theta \mapsto \theta + \pi$.

7 Conclusion & outlook

We promoted the critical-one picture to an operational framework with invariants (E, ϕ) , a continuous half-twist flow on $\mathbb{P}^1$, and unit-free protocols. **Next:** couple these to linear fractional dynamics (flows generated by 2×2 matrices), design noise-robust estimators for (E, ϕ) , and explore applications to *hybrid cognition & Möbius control* (Part IV).